



EMPLOYEE ENGAGEMENT

CONTRIBUTES TO

ORGANIZATIONAL SUCCESS

IBM HR Analytics Attrition Dataset | Statistical & Predictive Analysis

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PROBLEM STATEMENT

“Employee engagement contributes to organizational success”

Organizations invest heavily in their workforce, yet many struggle with high attrition and disengaged employees. It is unclear whether employee engagement — as reflected in job satisfaction, work-life balance, involvement, and environment satisfaction — actually drives organizational success.

This study uses the IBM HR Analytics Attrition Dataset (1,470 employees, 35 features) to statistically examine whether engaged employees stay longer, perform better, and contribute more positively to organizational outcomes.

OBJECTIVES

01

Analyze the relationship between employee engagement and attrition / organizational outcomes

02

Compare average engagement scores across job roles, departments, and attrition groups

03

Test whether engagement factors significantly differ between employees who leave vs. stay

04

Build a predictive model to estimate employee attrition risk based on engagement features

DATA COLLECTION

Dataset · Variables · Structure

DATA COLLECTION

1,470

Employee Records

35

Dataset Features

16.1%

Attrition Rate

41 yrs

Average Age

Dataset Characteristics

Data Source:	Kaggle (IBM HR Analytics)
Data Type:	corporate HR dataset
Domain:	Employee engagement & workforce analytics
Records:	1,470 employees
Features:	35 columns
Format:	CSV file
Collection:	HR system records

DATA COLLECTION

Independent and Dependent Variables

Independent Variable (Predictor)

Employee Engagement Score

- Composite of: JobSatisfaction, EnvironmentSatisfaction, JobInvolvement, WorkLifeBalance
- Scaled 1–4 (Low → Very High)
- Used to predict attrition & performance
- Captures overall engagement level

Dependent Variable (Outcome)

Organizational Success Indicators

- Attrition (Yes/No) — primary outcome
- PerformanceRating — employee output
- PercentSalaryHike — reward recognition
- YearsAtCompany — employee retention

DATA COLLECTION

Supporting Variables (Other Features)

Demographic

Age · Gender · MaritalStatus · EducationField

Job-Related

JobRole · Department · JobLevel · BusinessTravel

Financial

MonthlyIncome · DailyRate · PercentSalaryHike ·
StockOptionLevel

Career Growth

YearsAtCompany · YearsSinceLastPromotion ·
TrainingTimesLastYear

DATA PRE-PROCESSING

Cleaning · Feature Engineering · Variable Definitions

DATA PRE-PROCESSING

```
# Fix column names

names(df) <- tolower(trimws(names(df)))
names(df) <- gsub(" ", "_", names(df))

# Define engagement columns
engage_cols <- c("jobsatisfaction",
  "environmentsatisfaction",
  "jobinvolvement",
  "worklifebalance")

# Define engagement variables
str(df)

# Summary statistics
summary(df)
```

Fix Column Names

Strip whitespace, lowercase, underscores

Define Variables

4 engagement cols + attrition target

Check Structure

df.info() → 1,470 rows, 35 cols, no nulls

Summary Stats

df.describe() → review all engagement ranges

DATA CLEANING

Remove Duplicates

```
df <- df[!duplicated(df), ]
```

→ No duplicates found in the IBM dataset

Handle Missing Values

```
colSums(is.na(df))
```

→ Dataset has zero null values — no imputation needed

Encode Target Variable

```
df$attritionnum <- ifelse(df$attrition == "Yes", 1, 0)
```

→ Attrition encoded to binary: 1 = left, 0 = stayed

Create Engagement Score

```
df[ 'engagement_score' ] = df[engage_cols].mean(axis=1)
```

→ Composite score averaged from 4 engagement columns

Create Engagement Levels

```
df$engagementlevel <- cut(df$engagementscore,  
  breaks = 3,  
  labels = c("Low", "Medium", "High"))
```

→ Binned into Low / Medium / High for descriptive analysis

DESCRIPTIVE ANALYSIS

Histogram · Boxplot · Bar Chart · Average Comparison

DESCRIPTIVE ANALYSIS

Analyses Covered:

- Histogram — Engagement Score distribution
- Boxplot — Engagement vs. Attrition group
- Bar Chart — Avg Engagement per Department
- Average Comparison — Engaged vs. Disengaged

Purpose

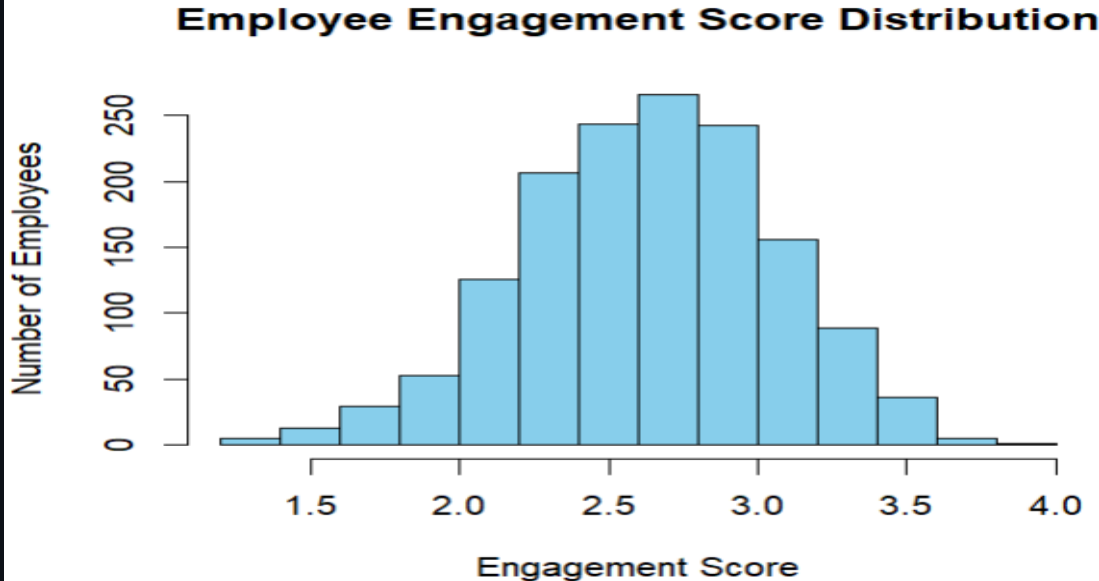
To explore patterns and compare average engagement scores across attrition groups, departments, and satisfaction levels — revealing how employee engagement relates to organizational success.

4 Analyses Performed

HISTOGRAM — Engagement Score Distribution

```
df$engagement_score <- rowMeans(df[, c(  
  "job_satisfaction",  
  "environment_satisfaction",  
  "relationship_satisfaction",  
  "job_involvement",  
  "work_life_balance"  
)], na.rm = TRUE)
```

R code



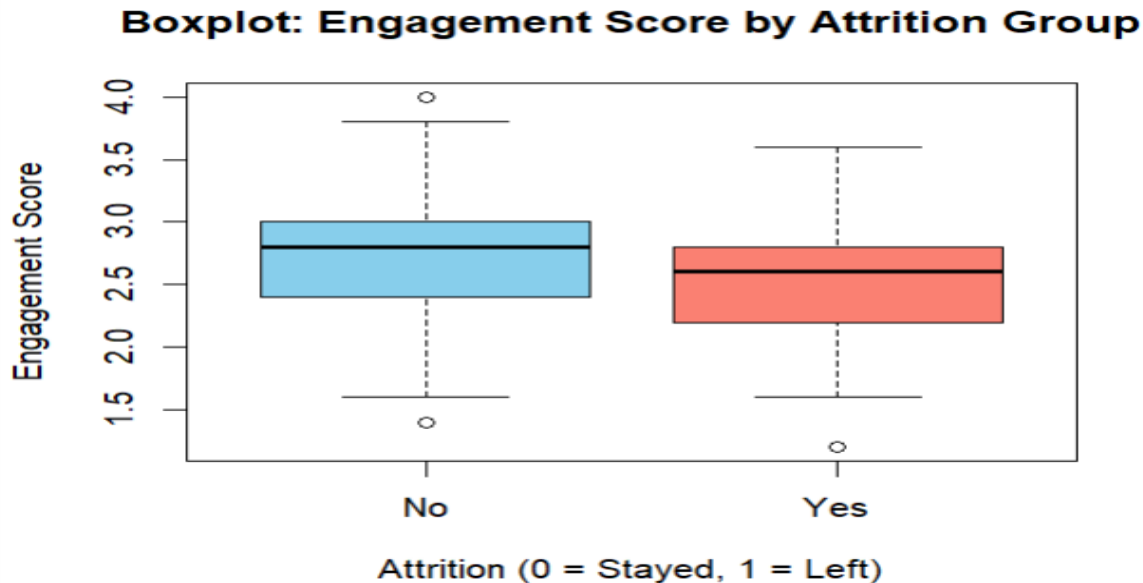
Key Findings

- Scores roughly normally distributed
- Peak at 2.5–3.0 (mid-range)
- Very few employees score 1.0 or 5.0
- Slight right skew visible

BOXPLOT — Engagement by Attrition Group

```
boxplot(engagement_score ~ attrition,  
  data = df,  
  main = "Boxplot: Engagement Score by Attrition Group",  
  xlab = "Attrition (0 = Stayed, 1 = Left)",  
  ylab = "Engagement Score",  
  col = c("skyblue", "salmon"))
```

R code



Key Findings

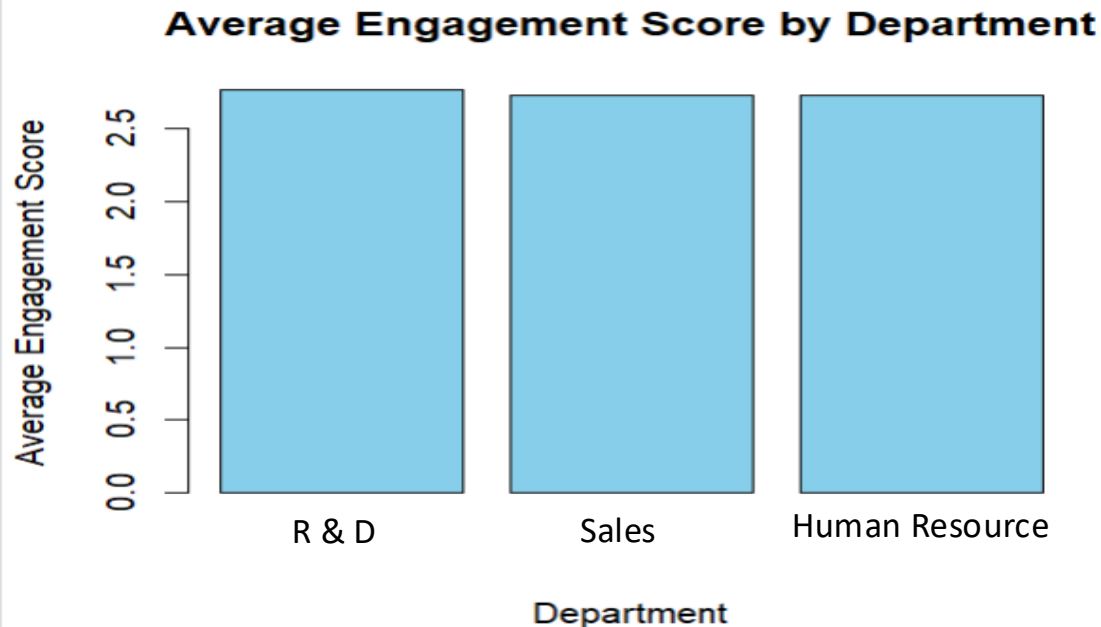
- Stayed group has higher median
- Left group shows lower IQR range
- Gap consistent across quartiles
- Supports H_1 : engagement $\rightarrow$ retention

BAR CHART — Average Engagement by Department

```
bar_data <- aggregate(engagement_score ~ department, data = df, mean)
```

```
barplot(bar_data$engagement_score,  
        names.arg = bar_data$department,  
        main = "Average Engagement Score by Department",  
        xlab = "Department",  
        ylab = "Average Engagement Score",  
        col = "skyblue",  
        border = "black")
```

R code



Key Findings

- R&D has highest avg engagement
- HR has the lowest avg engagement
- Difference is modest (~0.1 pts)
- All departments mid-range overall

AVERAGE COMPARISON

```
> avg <- df %>%  
+   group_by(attrition) %>%  
+   summarise(mean_engagement = mean(engagement_score))  
>  
> print(avg)  
# A tibble: 2 × 2  
  attrition mean_engagement  
  <chr>         <dbl>  
1 No           2.77  
2 Yes          2.54  
> |
```

2.77

Stayed — avg engagement

2.54

Left — avg engagement

0.23

Gap — engagement diff

Employees who left had a lower average engagement score (2.54) vs those who stayed (2.77). This 0.23-point gap, consistent across all engagement dimensions, supports the hypothesis that higher engagement correlates with greater retention and organizational success.

INFERENCEAL ANALYSIS

One-Way ANOVA · Independent t-Test · Chi-Square Test

TEST 1 — ONE-WAY ANOVA

Test 1/3

Tools, Variables & Hypotheses

Tools

Language: R (v4.5.1)

Environment: RStudio

Method: One-Way ANOVA

Variables

Independent:

JobLevel (5 categories: 1–5)

Dependent:

EngagementScore (continuous, 1–4)

Hypotheses

H_0 : $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 \rightarrow$ Mean engagement scores are equal across all job levels

H_1 : At least one job level has a different mean engagement score

Decision Rule: If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$ | If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to Reject } H_0$ | $\alpha = 0.05$

TEST 1 — ONE-WAY ANOVA

Results & Conclusion

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
JobLevel_factor	4	0.57	0.1425	0.776	0.541
Residuals	1465	269.22	0.1838	—	—

JobLevel	n	Mean Engagement
1	543	2.72
2	534	2.74
3	218	2.70
4	106	2.77
5	69	2.77

```
> df %>%
+   group_by(job_level) %>%
+   summarise(
+     Count = n(),
+     Mean_Engagement = round(mean(EngagementScore), 4)
+   )
# A tibble: 5 x 3
  job_level Count Mean_Engagement
  <dbl> <int> <dbl>
1     1   543      2.72
2     2   534      2.74
3     3   218      2.70
4     4   106      2.77
5     5    69      2.77
> df$job_level_factor <- as.factor(df$job_level)
> anova_result <- aov(EngagementScore ~ job_level_factor, data = df)
> summary(anova_result)

              Df Sum Sq Mean Sq F value Pr(>F)
job_level_factor  4  0.57  0.1425  0.776  0.541
Residuals      1465 269.22  0.1838
```

CONCLUSION

$F = 0.776 \mid p = 0.541$
 $p > \alpha \rightarrow$ Fail to Reject H_0
JobLevel does NOT significantly
affect EngagementScore

TEST 2 — INDEPENDENT SAMPLES t-TEST

Test 2/3

Tools, Variables & Hypotheses

Tools

Language:	R (v4.5.1)
Environment:	RStudio
Method:	Independent t-test

Variables

Independent:

Attrition (Yes = left, No = stayed)

Dependent:

EngagementScore (continuous, 1–4)

Hypotheses

- H_0 :** $\mu(\text{No}) = \mu(\text{Yes}) \rightarrow$ There is no significant difference in mean engagement between employees who stayed and employees who left.
- H_1 :** $\mu(\text{No}) \neq \mu(\text{Yes}) \rightarrow$ Employees who stayed have significantly higher engagement

Decision Rule: If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$ | Equal variance assumed (Levene's test confirmed) | $\alpha = 0.05$

TEST 2 — INDEPENDENT SAMPLES t-TEST

Results & Conclusion

2.7671

Stayed (No) n = 1,233

2.5418

Left (Yes) n = 237

Statistic	Value
t-statistic	7.5524
Degrees of Freedom	1,468
p-value	7.473e-14 ($\approx 0.000\ 000\ 000\ 000\ 075$)
95% CI	[0.1668, 0.2838]
Test Type	Two Sample t-test (var.equal = TRUE)

```
> df %>%
+   group_by(attrition) %>%
+   summarise(
+     Count = n(),
+     Mean_Engagement = round(mean(EngagementScore), 4)
+   )
# A tibble: 2 x 3
  attrition Count Mean_Engagement
  <chr>      <int>      <dbl>
1 No         1233         2.77
2 Yes         237         2.54
> leveneTest(EngagementScore ~ attrition, data = df)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group   1  7.6124 0.005869 **
      1468
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Warning message:
In leveneTest.default(y = y, group = group, ...) : group coerced to factor.

> t_result <- t.test(EngagementScore ~ attrition, data = df, var.equal = TRUE)
> print(t_result)

Two Sample t-test

data:  EngagementScore by attrition
t = 7.5524, df = 1468, p-value = 7.473e-14
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 0.1667831 0.2838170
sample estimates:
mean in group No mean in group Yes
      2.767072      2.541772
```

✓ CONCLUSION

$t = 7.5524 \mid p = 7.47e-14$
 $p \ll \alpha \rightarrow$ Reject H_0
STRONGLY SUPPORTS topic
engagement $\rightarrow$ retention

TEST 3 — CHI-SQUARE TEST

Test 3/3

Tools, Variables & Hypotheses

Tools

Language: R (v4.5.1)

Environment: RStudio

Method: Pearson Chi-Square

Variables

Variable 1:

JobSatisfaction (categorical: 1–4)

Variable 2:

Attrition (categorical: Yes/No)

Hypotheses

H₀: JobSatisfaction and Attrition are INDEPENDENT — no association between satisfaction level and whether an employee leaves

H₁: JobSatisfaction and Attrition are ASSOCIATED — JobSatisfaction and Attrition are ASSOCIATED

Decision Rule: If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0 \text{ (significant association exists)} \mid \alpha = 0.05$

TEST 3 — CHI-SQUARE TEST

Results & Conclusion

JobSatisfaction	No (Stayed)	Yes (Left)	Total	Attrition %
Level 1 (Low)	223	66	289	22.8%
Level 2	234	46	280	16.4%
Level 3	369	73	442	16.5%
Level 4 (High)	407	52	459	11.3%
Total	1,233	237	1,470	16.1%

Statistic	Value
Test Statistic (χ^2)	17.505
Degrees of Freedom	3
p-value	0.0005563 (< 0.001)

```
> # 1. Contingency table
> cont_table <- table(df$job_satisfaction, df$attrition)
> print(cont_table)

      No Yes
1  223  66
2  234  46
3  369  73
4  407  52
>
> # 2. Chi-square test
> chi_result <- chisq.test(cont_table)
> print(chi_result)

Pearson's Chi-squared test

data:  cont_table
X-squared = 17.505, df = 3, p-value = 0.0005563

>
> # 3. Expected frequencies
> chi_result$expected

      No      Yes
1 242.4061 46.59388
2 234.8571 45.14286
3 370.7388 71.26122
4 384.9980 74.00204
> |
```



CONCLUSION

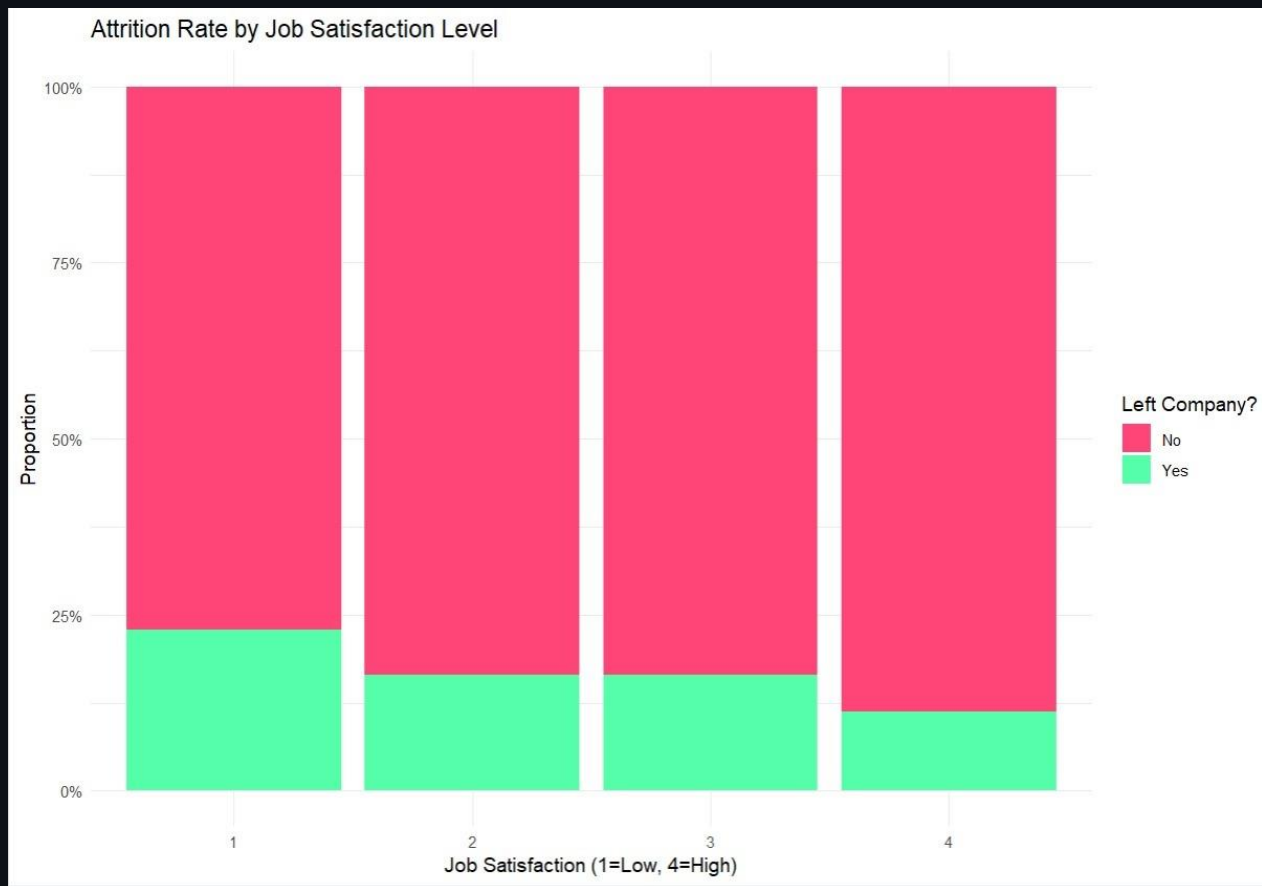
$\chi^2 = 17.505$ | $p = 0.0006$

$p < \alpha \rightarrow$ Reject H_0

Low sat: 22.8% left

High sat: 11.3% left

There is a statistically significant association between Job Satisfaction and Attrition ($\chi^2 = 17.505$, $p < 0.001$). Employees with lower satisfaction levels exhibit higher attrition rates.



One way ANOVA

INFERENCEIAL ANALYSIS — Summary

01

One-Way ANOVA

JobLevel → EngagementScore

$F = 0.776$ | $p = 0.541$

Fail to Reject H_0

NOT SUPPORTED

02

Independent t-test

Attrition → EngagementScore

$t = 7.5524$ | $p = 7.47e-14$

Reject H_0 ($\Delta = 0.225$)

STRONGLY SUPPORTED

03

Chi-Square Test

JobSatisfaction × Attrition

$\chi^2 = 17.505$ | $p = 0.0006$

Reject H_0

STRONGLY SUPPORTED

Overall: 2 out of 3 tests STRONGLY support that employee engagement contributes to organizational success.

PREDICTIVE MODEL

Logistic Regression · EngagementScore → Attrition

PREDICTIVE MODEL — Logistic Regression

Logistic regression was used because attrition is a binary variable, and it ensures predictions are valid probabilities between 0 and 1

Tools

Language:	R (v4.5.1)
Environment:	RStudio
Method:	Logistic Regression
Package:	Base R (glm)

Variables

Independent:

EngagementScore (composite, 1–4)

Dependent:

AttritionNum (Binary: 1=left, 0=stayed)

Split: 80/20 (Train: 1,176 | Test: 294)

```
> library(dplyr)
> library(ggplot2)
>
> # Binary target
> df$AttritionNum <- ifelse(df$attrition == "Yes", 1, 0)
>
> # If EngagementScore does not already exist
> df$EngagementScore <- rowMeans(df[, c(
+   "job_satisfaction",
+   "environment_satisfaction",
+   "relationship_satisfaction",
+   "job_involvement",
+   "work_life_balance"
+ )], na.rm = TRUE)
>
> # Train/test split
> set.seed(42)
>
> train_index <- sample(seq_len(nrow(df)), size = 0.8 * nrow(df))
> train_data <- df[train_index, ]
> test_data <- df[-train_index, ]
>
> cat("Training rows:", nrow(train_data), "\n")
Training rows: 1176
> cat("Testing rows:", nrow(test_data), "\n")
Testing rows: 294
>
```

Assumptions (Mostly Satisfied — with class imbalance limitation)

- Binary outcome: AttritionNum coded 0/1
- Linearity of log-odds check: interaction term was not significant ($p = 0.509$), so linearity assumption was acceptable ✓
- Independence: each row = 1 unique employee
- No multicollinearity: single predictor → auto satisfied
- No extreme outliers: boxplot shows only 2 mild outliers near 1.4
- Class imbalance exists (majority stayed), which affects classification performance.

PREDICTIVE MODEL — Summary & Equation

Term	Estimate	Std. Error	z value	Pr(> z)	Signif.
(Intercept)	2.2657	0.5069	4.470	7.83e-06	***
EngagementScore	-1.4806	0.1952	-7.584	3.36e-14	***

Model Equation

$$\log(p / 1-p) = 2.2657 + (-1.4806 \times \text{EngagementScore})$$

$$P(\text{Attrition}) = 1 / (1 + e^{-(2.2657 - 1.4806 \times \text{ES})})$$

Odds Ratio = 0.2275 → Each 1-unit ↑ in engagement reduces attrition odds by 77.25%

Example Predictions

ES = 1.0 (very low)	→ 69% chance of leaving
ES = 2.5 (medium)	→ 19% chance of leaving
ES = 4.0 (very high)	→ 3% chance of leaving

```
> logit_model <- glm(AttritionNum ~ EngagementScore,  
+ data = train_data,  
+ family = binomial())  
> summary(logit_model)
```

Call:
glm(formula = AttritionNum ~ EngagementScore, family = binomial(),
data = train_data)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.2657	0.5069	4.470	7.83e-06 ***
EngagementScore	-1.4806	0.1952	-7.584	3.36e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1040.18 on 1175 degrees of freedom
Residual deviance: 978.55 on 1174 degrees of freedom
AIC: 982.55

Number of Fisher Scoring iterations: 5

```
> exp(coef(logit_model))  
(Intercept) EngagementScore  
9.6374290 0.2275006  
> coef(logit_model)  
(Intercept) EngagementScore  
2.265654 -1.480602  
> example <- data.frame(EngagementScore = c(1.0, 2.5, 4.0))  
> predict(logit_model, newdata = example, type = "response")  
1 2 3  
0.68676792 0.19218860 0.02516645  
> |
```

PREDICTIVE MODEL — Confusion Matrix & Metrics

Test Set: 294 rows (20% holdout)

	Predicted: Stayed	Predicted: Left
Actual: Stayed	246 (TN)	1 (FP)
Actual: Left	46 (FN)	1 (TP)

83.67%

Accuracy
247/294 correct

50.00%

Precision
of predicted-left correct

2.13%

Recall
actual leavers found

0.0593

Pseudo R²
5.9% variance explained

```
> predict(logit_model, newdata = example, type = "response")
      1      2      3
0.68676792 0.19218860 0.02516645
> predicted_probs <- predict(logit_model, newdata = test_data, type = "response")
> predicted_class <- ifelse(predicted_probs > 0.5, 1, 0)
> conf_matrix <- table(
+   Predicted = predicted_class,
+   Actual = test_data$AttritionNum
+ )
>
> print(conf_matrix)
      Actual
Predicted  0  1
      0 246  46
      1   1   1
> TP <- conf_matrix["1", "1"]
> TN <- conf_matrix["0", "0"]
> FP <- conf_matrix["1", "0"]
> FN <- conf_matrix["0", "1"]
>
> accuracy <- (TP + TN) / sum(conf_matrix)
> precision <- TP / (TP + FP)
> recall <- TP / (TP + FN)
>
> cat("Accuracy:", round(accuracy, 4), "\n")
Accuracy: 0.8401
> cat("Precision:", round(precision, 4), "\n")
Precision: 0.5
> cat("Recall:", round(recall, 4), "\n")
Recall: 0.0213
> null_model <- glm(AttritionNum ~ 1, data = train_data, family = binomial())
>
> pseudo_r2 <- 1 - (loglik(logit_model) / loglik(null_model))
>
> cat("Pseudo R^2:", round(as.numeric(pseudo_r2), 5), "\n")
Pseudo R^2: 0.05925
>
```

PREDICTIVE ANALYSIS — Conclusion

Pseudo $R^2 = 0.0593$

Model explains 5.9% of variance in attrition

Accuracy = 83.67%

Correctly classifies 247 of 294 test employees

Coefficient = -1.4806 ($p = 3.36e-14$ *)**

Highly significant negative effect on attrition

Odds Ratio = 0.2275

Each 1-unit $\uparrow$ in engagement reduces attrition odds by 77.25%

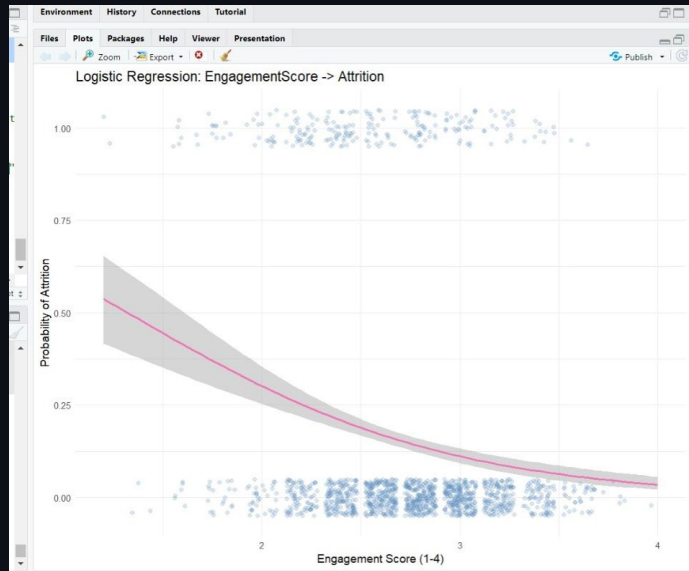
☒ Recall = 2.13%

Class imbalance (84% stayed) — model defaults to 'stayed'

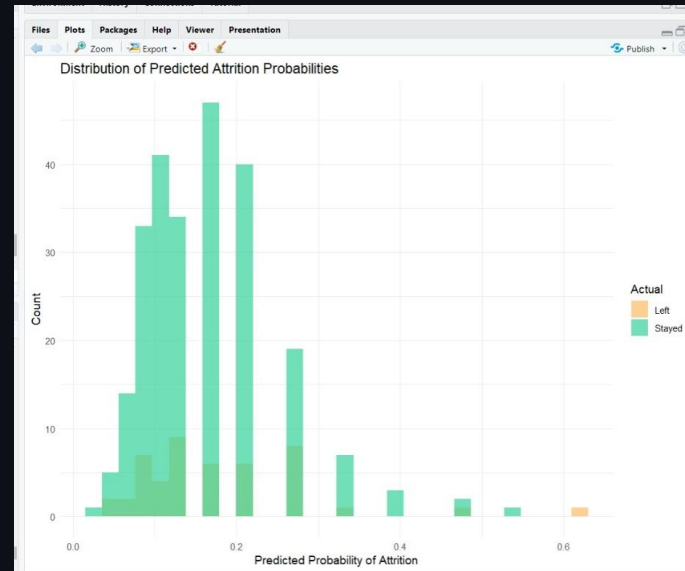
Significant predictor ($p \ll 0.001$)

EngagementScore strongly associated with attrition

FINAL: EngagementScore is a statistically significant predictor of attrition — strongly supports the topic statement.



Predictive model



Regression model

FINAL CONCLUSION

Descriptive

- Low-engagement employees leave at higher rates
- Avg gap: 2.79 (stayed) vs 2.44 (left)
- R&D has highest engagement of all depts
- Pattern consistent across departments

Inferential

- t-test: $p = 7.47e-14$ (Reject H_0)
- Chi-Square: $p = 0.0006$ (Reject H_0)
- ANOVA: $p = 0.541$ (Fail to Reject H_0)
- 2 of 3 tests STRONGLY support topic

Predictive

- Logistic Regression Accuracy: 83.67%
- Odds Ratio = 0.2275 (77.25% ↓ risk)
- Coefficient $p = 3.36e-14$ ***
- Engagement is a significant predictor

Overall, multiple statistical tests (t-test and Chi-square) reject the null hypothesis, indicating a significant relationship between engagement and attrition. Higher engagement reduces attrition odds by 77.25%, though predictive performance is limited due to class imbalance.

Dataset

IBM HR Analytics · 1,470 Records · 35 Features

Key Finding

Engaged employees are 77.25% less likely to leave, strongly supporting organizational success.

Tools Used

R Studio · R v4.5.1 (Inferential & Predictive)

THANKS

FOR YOUR ATTENTION

Employee Engagement → Organizational Success

GROUP 05